2026-10-13 | VS&A Daily |

When modes trade places



Ares I-X upper stage above the Vehicle Assembly Building access platforms. Photo: NASA, reproduced from Figure 10 of NASA/TM-2010-216182 [1]. Context only; the calculations here are not Ares test results.

Two bending directions. One stubborn correlation matrix.

Imagine a modal review in which the frequencies look right, the animations look familiar, and every entry in one two-by-two MAC block is 0.50. It is tempting to start adjusting stiffness. But that pattern can arise even when the two models span exactly the same motions.

The puzzle is especially relevant to nearly symmetric structures. NASA’s Ares I-X test report describes companion bending modes in orthogonal planes at nearly the same frequency. The first three free-free bending pairs were important to flight-control evaluations [1].

Today’s objective is to distinguish a change in the basis used to describe a mode cluster from a change in the physical motion that cluster can represent. We will also see why an excellent subspace comparison still cannot certify a frequency-response prediction.

Ares I-X makes the stakes concrete without supplying the numbers for our example. Its free-free vehicle test was impractical within project constraints, so tests during stacking and on the Mobile Launcher Platform helped calibrate the model [1]. Boundary conditions and modal identity had to be interpreted together.

The question for a review is therefore larger than “Which test mode matches mode 17?” Sometimes the more useful first question is “Do these two sets describe the same family of motion?” Only then should we ask whether the frequencies, damping and force participation make that family behave the same way.

A mode shape is a direction, not a serial number

For two nonzero real mode-shape vectors a and b, ordinary MAC is the squared cosine of the angle between them. It ignores sign and overall scale. This familiar collinearity test is useful precisely because mode-shape amplitude is arbitrary [2].

$$\mathrm{MAC}(\mathbf{a},\mathbf{b}) = \frac{|\mathbf{a}^T\mathbf{b}|^2}{(\mathbf{a}^T\mathbf{a})(\mathbf{b}^T\mathbf{b})}$$

The vectors must describe the same coordinates and directions. For complex shapes, use conjugate transpose instead of transpose. Here we use real shapes, equal coordinate weighting and a linear system; no claim about complex-mode validation is needed.

Now take a mass-normalized structure with an exactly repeated natural frequency. Every linear combination of the two eigenvectors remains an eigenvector at that frequency. The eigensolver can return one orthonormal pair, while a second solution returns a rotated pair. Neither basis has a privileged compass direction.

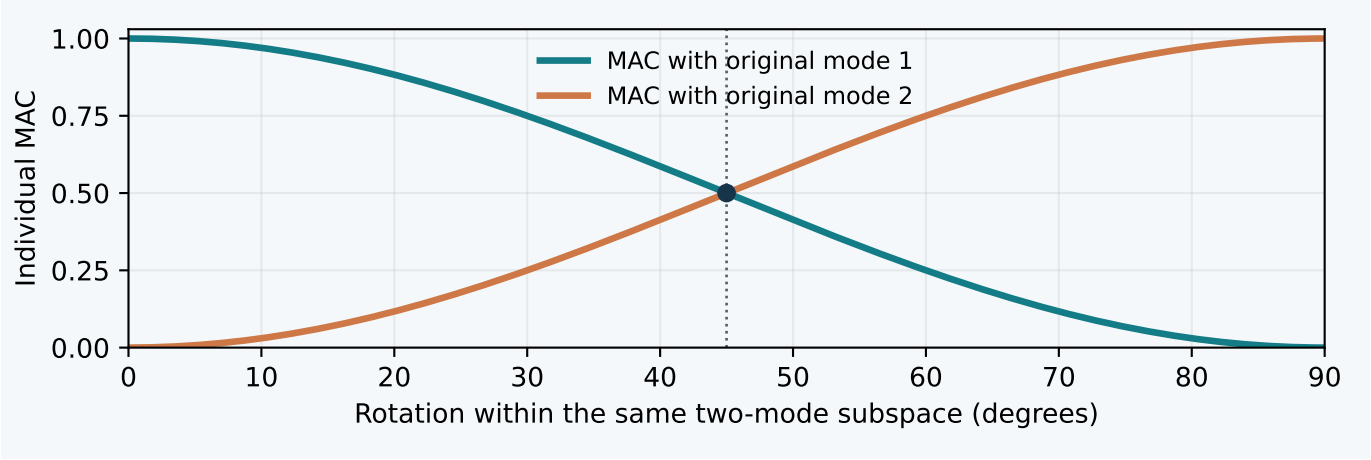


Figure 1. Original analytical plot: rotating one shape between two orthonormal directions transfers its individual MAC from one partner to the other. At 45°, both scores are 0.50, although the complete two-dimensional subspace is unchanged.

Close but unequal frequencies require more care. Small asymmetries or boundary changes can select very different preferred directions, and mode order can exchange. Arbitrary rotation is no longer an exact eigenvector freedom. The cluster comparison remains informative, but the distinct modal dynamics must remain attached to their shapes.

Garvey, Penny and Friswell explicitly separated subspace mismatch from alignment within a common subspace in their 1996 correlation study [3]. Those are different errors, and a single pairwise score can mix them together.

A 0.50 MAC with no missing motion

Use three equally weighted comparison coordinates. Let the analytical pair be A = [e_1, e_2], where e_1 = (1, 0, 0) and e_2 = (0, 1, 0). The unused third direction leaves room to detect a genuine mismatch. Define a second orthonormal pair:

$$\mathbf{b}_1 = \frac{\mathbf{e}_1 + \mathbf{e}_2}{\sqrt{2}}, \qquad \mathbf{b}_2 = \frac{-\mathbf{e}_1 + \mathbf{e}_2}{\sqrt{2}}$$

Each dot product with either analytical direction has magnitude $1/\sqrt{2}$. Squaring gives 0.50 in all four MAC cells. Reordering the columns cannot produce a strong diagonal. Yet any displacement in the analytical plane can be reconstructed exactly from the second pair.

To compare the planes, first orthonormalize both sets on the common coordinates, obtaining Q_A and Q_B. The singular values of their cross-product are cosines of the principal angles:

$$\sigma_i = \mathrm{svd}_i\left(Q_A^T Q_B\right) = \cos\theta_i$$

For this rotation, both singular values are 1, so both principal angles are zero. The largest angle is a useful diagnostic: it asks how poorly the least-aligned direction in one subspace can be represented by the other.

As a control case, replace the second test vector with $\cos(20°)e_2 + \sin(20°)e_3$ and retain e_1 as the first. The principal angles become 0° and 20°. One direction is now genuinely outside the analytical plane. This is not merely a relabeling.

Illustrative comparison	Pairwise MAC	Principal angles
45° rotation in the same plane	All four: 0.500	0°, 0°
One direction tilted out of plane	Diagonal: 1.000, 0.883	0°, 20°

Table 1. Original exact-vector calculations, not measured data. MAC = $\cos^2(20°)$ = 0.883 for the tilted direction. Equal subspace dimension and independent columns are essential; adding extra comparison modes can make containment easier without proving equivalence.

The catch: the same plane can move differently

Subspace agreement removes one ambiguity. It does not remove the poles. Consider two mass-normalized modes at 50 and 52 Hz, each with 1% viscous damping. Apply and observe force and displacement along e_1. In the original basis, only the 50 Hz mode contributes to this receptance.

$$h_r(\omega) = \frac{1}{\omega_r^2 - \omega^2 + 2i\zeta_r\omega_r\omega}$$

Here $\omega = 2\pi f$ is angular frequency in rad/s, ζ is damping ratio, and h is the modal receptance for unit modal mass. Rotate the shapes 45° but continue assigning the two original poles to those rotated shapes. The directional responses are then:

$$H_{xx}^{\text{original}} = h_1, \qquad H_{xx}^{\text{rotated}} = \frac{h_1 + h_2}{2}$$

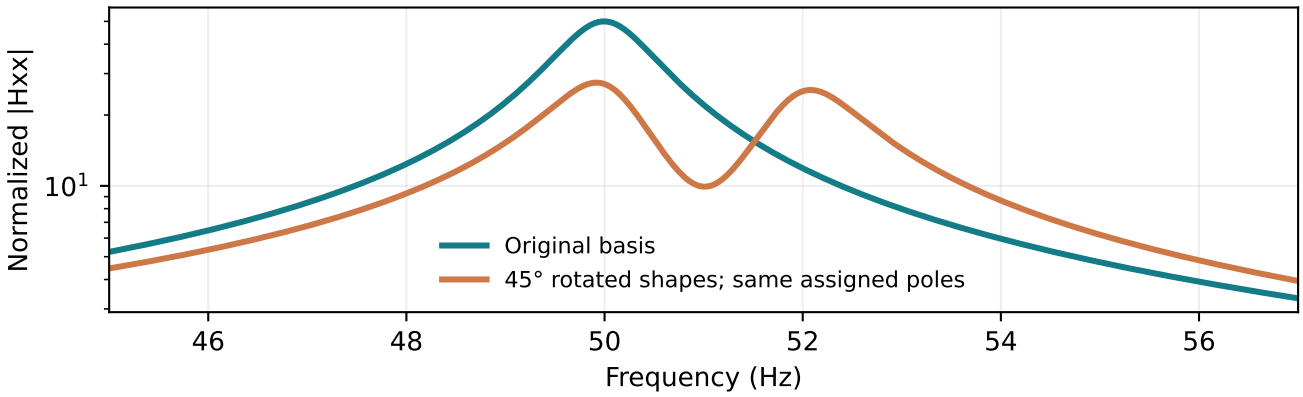


Figure 2. Original simulated receptances, multiplied by ω_1^2 to normalize the unit-modal-mass response. Both systems occupy exactly the same modal plane, but the second has different stiffness orientation. They are physically different systems.

At 50 Hz, the normalized magnitude falls from 50.0 to about 27.1, a ratio of 0.542. The principal angles remain zero. A perfect subspace score has missed a substantial directional response difference because it contains no frequency or damping information.

For exactly equal frequencies and equal modal damping, $h_1 = h_2$ and the rotation leaves the complete pair's response unchanged. For unequal poles, a true coordinate transformation also transforms the modal stiffness and damping matrices, generally making them nondiagonal. Keeping diagonal poles while rotating only the shapes changes the model.

This is why the diagnostic belongs beside measured frequency-response functions, residues and the intended loads or control outputs. It can explain poor pairing; it cannot excuse poor response prediction. The numerical example isolates that distinction and does not establish an acceptance tolerance.

What to bring back to the review

Start with coordinate mapping, sensor directions, boundary conditions and independent identification of the full cluster. Then show pairwise MAC, the cluster's principal angles, frequency splitting and damping together. Check repeated runs before interpreting a rotated direction as a hardware change.

Choose the comparison metric deliberately. Euclidean weighting describes the measured coordinate space. A positive-definite mass weighting can describe a different, energy-related geometry, but requires a defensible mass representation and consistent reduction or expansion. Neither approach validates motion at unmeasured coordinates automatically.

Do not improve the score by silently enlarging the cluster, discarding a weakly observed mode or rotating shapes while leaving their dynamics behind. Preserve rank information and judge the effect on the actual force-to-response paths that drive the decision.

NASA-STD-5002B, approved March 10, 2026, requires mass-weighted cross-orthogonality checks and specified correlation criteria for applicable spacecraft and payload models. LAR 64 calls for assessing effects on critical loads when listed criteria are not met; LAR 65 ties model adjustments to as-built hardware. Its scope excludes launch vehicles and aircraft. A subspace score is supporting evidence, not an automatic substitute for applicable requirements [4].

Three takeaways

1. Poor individual MAC can reflect basis rotation within a well-matched mode cluster.
2. Principal angles expose missing directions, but do not test frequencies, damping or residues.
3. Accept a model for its intended predictions using the governing criteria and physically justified changes.

Reflection. A close pair has zero principal angles but a poor measured FRF match. Is correlation complete?
Answer. No. Inspect the pole splitting, damping, modal scaling and input/output participation; the span alone cannot determine the response.

Sources and figure credits

[1] R. D. Buehrle et al., Ares I-X Flight Test Vehicle Modal Test, NASA/TM-2010-216182, January 2010, sections 1-2 and Figure 10. NASA report and photograph.

[2] Structural Vibration Solutions, ARTeMIS Modal documentation, "Modal Assurance Criterion," undated; accessed October 13, 2026. Definition and estimation uncertainty.

[3] S. D. Garvey, J. E. T. Penny and M. I. Friswell, "Quantifying the Correlation Between Measured and Computed Mode Shapes," Journal of Vibration and Control 2(2), 123-144, April 1996. DOI and public abstract. Cited for its error taxonomy; worked examples are original.

[4] NASA-STD-5002B, Load Analyses of Spacecraft and Payloads, March 10, 2026, sections 1.2 and 4.9.2. Official standard. Figures 1-2 and Table 1: original analytical illustrations.